

OptArrow Technical Specification

Introduction

OptArrow serves as a high-performance optimization integration engine, focusing on bridging optimization clients with solver backends through an efficient transport layer. This specification, constrained to verified claims, outlines its architectural components, data interchange mechanisms, runtime environments, and licensing details.

1. Engine Architecture

1.1 Python Engine Adaptor and Flight Server

Architecture Alignment

OptArrow employs a Python Engine Adaptor alongside a Flight Server to manage data communication and transformation:

- **Python Engine Adaptor:** Facilitates initial conversion of optimization problem data from native Python structures (e.g., Dictionaries) into an intermediate format suitable for transport and processing.
- **Flight Server:** Utilizes Apache Arrow Flight to provide a transport layer that efficiently manages data streams between components, enabling the platform-independent communication required for optimization tasks.

Interface Layers

In addition to the core components, OptArrow includes interface layers for Julia and MATLAB, allowing seamless cross-language operations:

- **Julia Interface Layer:** Captures Arrow IPC data directly via Julia's native inter-process capabilities, converting it back to suitable Julia data structures for further computational processing.
- **MATLAB Interface:** Provides a lightweight client to interact with the core engine, enabling users from MATLAB environments to handle optimization tasks via OptArrow's infrastructure.

2. Functional Capabilities

2.1 Linear Programming (LP) and Quadratic Programming (QP)

Solver Backend Integration

OptArrow's primary functional capabilities include solving LP and QP problems through integrated solver backends such as the HiGHS library:

- **HiGHS Integration:** Directly processes Arrow IPC data into the HiGHS API format to execute LP and QP computations.
- **Result Handling:** Post-computation, results are converted from HiGHS specific formats

back to Arrow IPC, further transformed into Python Dicts or Julia native structures depending on the client environment.

3. Technology Integration

3.1 Apache Arrow Data Interchange

Mechanisms and Interfaces

OptArrow leverages Apache Arrow to enable efficient data interchange between Python and Julia:

- **Data Transformation Workflow:**
 - **Input Stage:** Python Dict representing optimization problem.
 - **Arrow Conversion:** Transforms Python Dict to Arrow format using zero-copy in-memory structures.
 - **Julia Integration:** Direct access to Arrow IPC, used for computation by interfacing with Julia's data structures.
- **Performance Consideration:** Arrow's design ensures minimal overhead, thus supporting high-throughput, real-time computations suited for large optimization problems.

4. Runtime Environments

4.1 Supported and Extended Environments

Primary Support

OptArrow is engineered to operate within Python and Julia environments:

- **Python Workflow:** Primary scripting and processing environment, leveraging robust libraries and built-ins to facilitate initial data handling and API calls to OptArrow.
- **Julia Processing:** Utilizes Julia's performance advantages for numerical computations directly interacting with Arrow data.

Extensions: MATLAB Integration

The MATLAB interface provides additional access:

- **Functionality Extension:** MATLAB users interact through a lightweight API that abstracts complex inter-language operations, offering users an accessible bridge into OptArrow's core capabilities.

5. Software Licensing and Community Alignment

5.1 Licensing Details

Open-Source Framework

OptArrow's licensing strategy supports open-source and community-driven software development:

- **Apache License 2.0:** Covers substantial components, promoting adaptability and extensive use.
- **Creative Commons Attribution 4.0:** Ensures open-source content and educational resources are freely available.

Together, these licenses support a collaborative environment fostering innovation and community engagement in optimizing solutions and research advancements.

Conclusion

This specification confirms that OptArrow integrates cross-language capabilities, high-efficiency transport, and security mechanisms, primarily tailored for solving LP and QP problems. It supports innovation through community-aligned licensing, providing a robust, adaptable platform for optimization challenges in various computational environments.